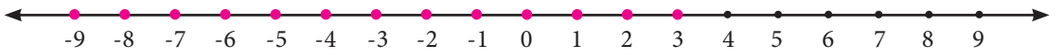
$$\begin{aligned} 4y &\leq 12 \\ \frac{4y}{4} &\leq \frac{12}{4} \quad \text{[Dividing by 4 on both sides]} \\ y &\leq 3 \end{aligned}$$

Since the solution belongs to integers, the solutions are 3, 2, 1, 0, –1, –2,..... It's graph on the number line is shown below:



Exercise 3.2

1. Given that $x \geq y$. Fill in the blanks with suitable inequality signs.

   (i) y ☐ x

   (ii) $x + 6$ ☐ $y + 6$

   (iii) x^2 ☐ xy

   (iv) $-xy$ ☐ $-y^2$

   (v) $x-y$ ☐ 0

2. Say True or False.

   (i) Linear inequation has almost one solution.

   (ii) When x is an integer, the solutions for $x \leq 0$ are $-1, -2,...$

   (iii) An inequation, $-3 < x < -1$, where x is an integer, cannot be represented in the number line.

   (iv) $x < -y$ can be rewritten as $-y < x$

3. Solve the following inequations.

   (i) $x \leq 7$, where x is a natural number.

   (ii) $x - 6 < 1$, where x is a natural number.

   (iii) $2a + 3 \leq 13$, where a is a whole number.

   (iv) $6x - 7 \geq 35$, where x is an integer.

   (v) $4x - 9 > -33$, where x is a negative integer.

4. Solve the following inequations and represent the solution on the number line:

   (i) $k > -5$, k is an integer.

   (ii) $-7 \leq y$, y is a negative integer.

   (iii) $-4 \leq x \leq 8$, x is a natural number

   (iv) $3m - 5 \leq 2m + 1$, m is an integer.

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